

Zero-Shot NER in Slavic Languages: The Impact of Linguistic Relations, Lemmatization, and Transliteration

Dimitar Kochev Mariia Zviahintseva Mihael Stoyanov Tetiana Tretiak

IT University of Copenhagen, Denmark

diko@itu.dk, mazv@itu.dk, svst@itu.dk, tetr@itu.dk

Abstract

In this project, we explore how to improve cross-lingual named entity recognition (NER) for Slavic languages. Our idea is that the linguistic relation could be utilized in building good models with fewer resources. We focus on performing zero-shot learning experiments between four languages - Russian, Bulgarian, Ukrainian, and Slovene. To address challenges related to alphabet differences and rich morphology, we apply two preprocessing techniques: transliteration of Slovene from Latin to Cyrillic script and lemmatization across all languages, then we use the XLM-R model configuration to build our models. We hypothesize that (H1) leveraging linguistic similarities, (H2) transliteration, and (H3) lemmatization can enhance model robustness and facilitate generalization to unseen Slavic languages. Our final results support H1, achieving a best F1 score of 89% for a Bulgarian model predicting Slovene and 82% for a Russian model predicting Ukrainian. However, H2 is not supported, and there is no conclusive evidence to either confirm or reject H3.

Our project repository: [GitHub](#)

1 Introduction

Named Entity Recognition (NER) is one of the fundamental tasks in natural language processing that involves identifying and classifying named entities such as people, organizations, and locations in the text. While current NER models have achieved significant accuracy in high-resource languages (e.g., English or Russian), performance often decreases significantly for low-resourced languages, such as many in the Slavic language group, e.g., Slovene and Ukrainian.

To address this problem, researchers started implementing the so-called cross-lingual zero-shot learning method, which involves training a model on one or more high-resourced languages and testing directly on data from a low-resource target lan-

guage. This can help reduce the cost and effort of data annotation.

Languages from the Slavic family are an interesting case for this. They are linguistically related and have some fundamental similarities in grammar and vocabulary. However, they have some significant differences, such as the scripts they use. Ukrainian, Russian, and Bulgarian use the Cyrillic alphabet, while Slovene uses the Latin script. In this case, transliteration can potentially improve model alignment across languages.

Another challenge is the morphological complexity. Many Slavic languages are highly inflected, meaning a single word can appear in many forms depending on case, gender, and number, which can confuse NER models. To overcome this, lemmatization, a process that reduces words to their dictionary (base) forms, can be applied.

Our research focuses on finding an efficient way to build a reliable NER model for low-resourced languages. For this purpose, we aim to improve the cross-lingual NER performance for Slavic languages by exploring the combined effects of transliteration, lemmatization, and linguistic similarity. Therefore, we chose to focus on four Slavic languages: Russian and Ukrainian from the East Slavic group, and Bulgarian and Slovene from the South Slavic group. We choose them as they belong to two distinct language sub-families and have strong linguistic ties within each group. Our four models are based on the XLM-R-large configuration in HuggingFace: one trained only on Bulgarian data (bg-ner), one only on Russian data (ru-ner), one with Bulgarian and Russian (bg+ru-ner), and one on all four languages. Based on the linguistic similarities, we expect that a model fine-tuned on Russian will perform better on Ukrainian data, while a model fine-tuned on Bulgarian will perform better on Slovene. The analysis of our results supports some of our linguistic assumptions.

2 Related Work

Most current studies focus on architecture-level improvements or large-scale multilingual training, with less attention to linguistic relationships within Slavic subgroups. While cross-lingual NER has been widely studied, we found little work that explicitly exploits linguistic similarities between Slavic languages. Current state-of-the-art systems often rely on pre-trained models (e.g., DeepPavlov’s multilingual BERT, WikiNEuRal), and some incorporate weak supervision and silver data (Piskorski et al., 2024) to overcome the lack of annotated data.

Our project takes a new approach by focusing on language-specific preprocessing to improve zero-shot NER. We explore how transliteration, lemmatization, and grouping languages by topology (East vs. South Slavic) affect model performance are aiming to identify factors that can enhance NER performance for low-resource Slavic languages.

As discussed in (Pałka and Nowakowski, 2023), lemmatization can be particularly important for languages with rich inflectional morphology. For instance, in Russian, a name such as Иван (Ivan) can appear with various suffixes (-a, -e, -ом, -y), depending on grammatical case. In contrast, the same name remains unchanged in Bulgarian. Consequently, if a model is trained on a language like Bulgarian and tested on Russian, the absence of lemmatization may lead to misclassifications or missed entities due to inflectional variation. To mitigate this, we hypothesized that applying lemmatization can improve the NER tagging objective, so we applied the Stanza lemmatizer uniformly across all data before training our models and conducted tests.

For the choice of model configuration, prior work on cross-lingual NER (e.g., (Ulčar et al., 2021), (Viksna et al., 2023), (Arkhipov et al., 2019), (Pałka and Nowakowski, 2023)) has compared multilingual transformer models such as mBERT, XLM-R, and Slavic-BERT. In (Ulčar et al., 2021), XLM-R outperforms mBERT in zero-shot NER between Slovenian and Croatian, with F1 scores of 0.769 vs. 0.751 and 0.882 vs. 0.844, respectively. This pattern holds across other Slavic languages as well, as shown in (Ferreira et al., 2021), and is consistent with findings from SlavNER shared tasks (Piskorski et al., 2021), which highlight performance variation due to script, mor-

phology, and resource differences. We therefore opted for XLM-R large (Conneau et al., 2020), as it consistently matches or exceeds mBERT, offering more reliable tokenization for our multilingual NER setup.

3 Dataset

For this project, we use the dataset (3rd Shared Task on SlavNER) provided by the BSNLP 2021 Shared Task organizers to train and evaluate our models. The data consists of news articles on different topics collected from web sources. The annotation files follow the Universal NER convention. The dataset includes text in six Slavic languages: Ukrainian (uk), Slovene (sl), Russian (ru), Polish (pl), Czech (cs), and Bulgarian (bg). Five named entity categories are used in the annotation: PER (person), LOC (location), ORG (organization), EVT (event), and PRO (product). We focus on four languages:

Language	Train	Validation	Test
Russian (ru)	7,381	5,045	10,486
Bulgarian (bg)	11,715	3,104	3,384
Ukrainian (uk)	2,211	1,850	1,493
Slovene (sl)	7,512	5,186	4,312

Table 1: Sentence counts per language and dataset split

4 Experimental setup

4.1 Linguistic Similarity

The language classification provided by Glottolog (Glottolog, 2024) offers insight into the relationships among various languages. The Slavic language family falls within the broader Indo-European family, specifically under the Classical Indo-European branch. It is further divided into three subgroups: East, West, and South Slavic. Our research focuses exclusively on the South and East Slavic languages. Russian and Ukrainian belong to the East Slavic subgroup, whereas Bulgarian and Slovene are part of the South Slavic subgroup. This classification suggests that Russian and Ukrainian are more closely related to each other than to Bulgarian or Slovene, and vice versa. For humans, this linguistic proximity often corresponds to higher mutual intelligibility between more closely related languages. Based on this, we hypothesize that such language similarity can be exploited in cross-lingual NER tagging. Specifi-

cally, we expect that a model trained on Bulgarian will perform better on Slovene than a model trained on Russian.

4.2 Writing Systems

As previously mentioned, one of the languages we focus on in our research uses a Latin-based script, while the other three use Cyrillic. To explore whether script alignment could enhance Slovene’s prediction accuracy, we investigated transliteration’s impact by converting Slovene data into Cyrillic. We hypothesize that transliterated Slovene will yield better results due to the alignment with the Cyrillic script used by the training languages.

To be more specific, since Bulgarian is linguistically the closest to Slovene, we chose to use a Bulgarian-based version of the Cyrillic alphabet. The transliteration was implemented using a custom-built tool, as unlike from Cyrillic to Latin, there is no built-in tool that we could use. Most Latin letters have direct Cyrillic equivalents (e.g., $b \rightarrow \text{б}$), but the letter j posed a challenge. Its representation varies by context: it can correspond to й or ь , or combine with certain vowels to form new characters. For instance, when j follows l or n , it softens the consonant (e.g., $lj \rightarrow \text{лъ}$, $nj \rightarrow \text{нъ}$). When paired with vowels like a or u , it becomes я (ja) or ю (ju), respectively. In all other cases, it is transliterated as й . Our code first handles the specific cases of j and then maps the remaining letters to their direct equivalent.

4.3 Model selection

In our Linguistic analysis section, we discussed the relation between the major Slavic groups (East and South Slavic). To evaluate the significance of their linguistic proximity, we decided to select one representative of each group (Bulgarian and Russian), fine-tune a model, and perform zero-shot testing. For example, we expected that our Bulgarian model would perform better compared to our Russian model when testing on Slovene data. The same strategy was used when evaluating on Ukrainian data, with the Russian model scoring higher.

Some of the settings in our experiments and some of the code we developed were inspired by the following notebook (Paul, 2023). More specifically, we used the code structure for the model building and the idea for comparing our model performance to simple models trained with a smaller

number of samples of the target low-resource language.

To build our NER tagging system, we use the body of the XLM-R configuration and attach a customized head that can perform token classification. We also use the default AdamW optimizer, since it maintains steady performance when handling large models like the Transformers.

Furthermore, we created models trained on combined data. One of those models was trained on a combination of our Bulgarian and Russian data. The other one - a more diverse model with all of our available datasets. The idea was to replicate the state-of-the-art concepts and use them as a benchmark to evaluate our zero-shot.

After developing our initial four models, we recognized the need to assess the impact of lemmatization on performance. As shown in the study by (May et al., 2019), lemmatization may influence NER accuracy, but this effect should be empirically verified rather than assumed. To investigate this, we trained an additional model using Russian data without lemmatization. We did not conduct a similar experiment for Bulgarian, as its minimal grammatical inflection reduces the potential impact of lemmatization on model performance.

5 Experiment Details

To perform the transformations mentioned in our previous sections, we constructed the following pipeline that has all of the steps required to prepare our data for training:

- **Lemmatization** – applied to each language file.
- **Transliteration** – We apply this only to the lemmatized Slovene data.
- **Tokenization** – our tokenizer is the pre-trained tokenizer from XLM-R-large. It has around 250,000 tokens generated after analyzing multilingual data. This is perfect for our settings because it can cover both Cyrillic and Latin scripts. It has also seen previous examples of our target languages, which makes the tokenization of the data reliable.

6 Results

After training the bg-ner , ru-ner , bg+ru-ner , and all-langs-ner models, we evaluated them on the available test sets for all four languages, as well

Evaluated on	ru	bg	uk	sl	sl_trans	Average
Fine-tune on						
ru	0,781	0,852	0,825	0,861	0,749	0,814
bg	0,754	0,870	0,774	0,892	0,719	0,802
ru+bg	0,780	0,865	0,811	0,896	0,723	0,815
all	0,775	0,890	0,812	0,883	0,761	0,824

Table 2: Span F1 results of our models

as on the transliterated version of Slovene. Table 2 presents the performance of each model on each language, along with the average performance across all languages. We highlighted the performance of our all-langs-ner model, which serves as the baseline for comparison. This allows us to assess whether training on only linguistically related languages can match - or even outperform - the larger multilingual model when applied to a specific target language. We also analyzed the results with respect to each identified modifier. As a performance metric, we decided to use the normal span F1 score.

6.1 Effect of Linguistic Similarity

As shown in Table 2, the model trained on Bulgarian performed better on Slovene, achieving a span F1 score of 89.2%, compared to 86.1% for the Russian-trained model and 88.3% for our generalized model. Similarly, for Ukrainian, the Russian model achieved a higher score of 82.5%, while the Bulgarian model reached only 77.4% and the generalized model score was 81.2%.

6.2 Effect of Transliteration

Latin Tokens	Cyrillic Tokens
[_krist, jan, ka]	[_кри, ст, й, а, нка]
[_ob, to, ž, en, kin]	[_об, то, жең, кин]
[_njen]	[_њ, ен]
[_števil, en]	[_шт, еви, лен]
[_nas, prot, ovanje]	[_нас, про, това, нь, е]
[_bo, go, klet, stvo]	[_бого, клет, ство]

Table 3: Comparison of Latin and Cyrillic Tokens

On the other hand, we observed that transliterating Slovene into Cyrillic led to a decrease in model performance. This is likely due to the fact that the transliterated words are not properly recognized by the XLM-R-large tokenizer, resulting in suboptimal tokenization and, consequently, less accurate NER predictions. As shown in Table 3, there is a significant disparity between the Latin and Cyrillic tokens. In the case of Cyrillicized Slovene, words

are often split into a greater number of tokens, many of which do not represent meaningful sub-word units. In contrast, all trained models achieved high accuracy on the original, non-transliterated Slovene data.

6.3 Effect of Lemmatization

We trained and tested the Russian model using both lemmatized and non-lemmatized data. The model with lemmatized data yielded a span F1 of 78.1%, whereas the model with non-lemmatized data reached 78.2%.

6.4 Effect of adding other languages

According to our results, the model trained on all languages achieved the highest average performance, with a span F1 score of 82.4%. However, a closer analysis of individual languages reveals that this model underperforms compared to the bg-ner and ru+bg-ner models on Slovene, and to the ru-ner model on Ukrainian. Similarly, while the ru+bg-ner model outperformed both ru-ner and bg-ner in terms of average performance, the margin over bg-ner on Slovene is minimal (only 0.004), and ru-ner still yields the better results on Ukrainian.

6.5 Effect of zero-shot learning

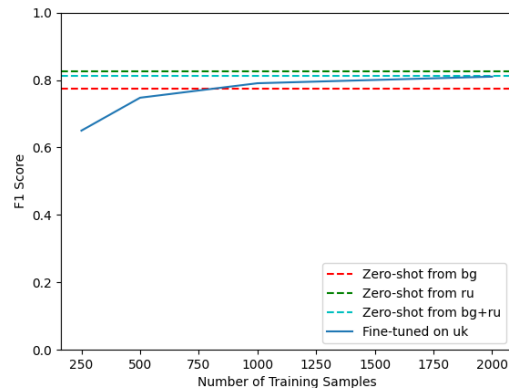


Figure 1: Ukrainian

Figure 1 shows that the zero-shot Russian model outperforms the model fine-tuned on a small Ukrainian dataset. At least 2,000 Ukrainian training samples are needed to reach a comparable F1 score, which can be challenging and time-consuming to obtain.

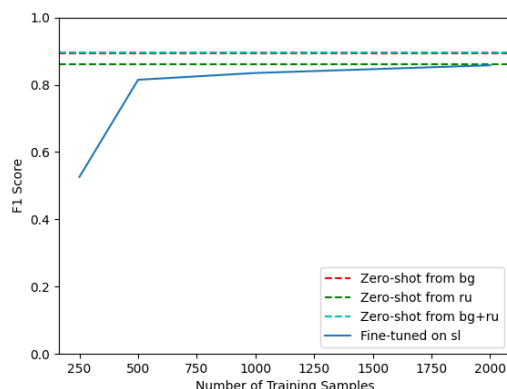


Figure 2: Slovenian

A similar trend is observed for Slovene, where the zero-shot model trained on Bulgarian already achieves high performance, and further fine-tuning on Slovene data provides only marginal gains.

7 Conclusion

In our project, we explored how linguistic relationships within the Slavic language family can enhance NER performance for low-resourced languages. To tackle this challenge, we utilized zero-shot learning, aiming to identify which high-resource languages yield the best accuracy on test sets from other Slavic languages and to determine which factors contribute to improved performance.

For each model, we built a pipeline that processes training and validation datasets for the respective language. This includes lemmatization of the data, applying transliteration specifically for Slovene, and tokenizing the text. Then we proceed with training the model using the XLM-R-large configuration.

Based on our results, we can conclude that, given the modest performance gains, zero-shot learning remains a solid and practical solution for low-resource languages. The findings also support our H1 regarding linguistic similarity, confirming that zero-shot NER performance improves when the training and the target languages are more closely related. Regarding the transliteration of Slovene, we reject the H2 hypothesis that it improves NER prediction in a cross-lingual zero-shot setting. The results further demonstrate that lemmatization does not have a noticeable effect on the performance, so we do not have enough evidence to confirm or deny our H3 about it.

In addition, training a model on a combination of languages proved to have high robustness

and, therefore, led to better average performance across all four target languages. However, this improvement does not necessarily translate to better performance in each individual language. In most cases, language-specific or bilingual models outperformed the multilingual one. So, increasing the size and diversity of the training data can boost generalization, but for optimal performance on specific languages, the linguistic relevance of the training data may be more important than its sheer size.

7.1 Limitations

- Our dataset is derived from news articles across four topics. Although these articles use formal language, they are non-academic in nature. To obtain more broadly applicable results, we should expand the dataset to include a wider range of text types, such as informal and academic texts from various domains.
- For Slovene, transliteration could be beneficial for multilingual models, provided we have a pretrained tokenizer that can correctly tokenize transliterated words. Alternatively, with significantly more data, we could train a tokenizer ourselves. Yet this would make the solution useless for applications outside research, since a Cyrillic version of Slovene would never be used in the real world.
- We could further explore the language similarity hypothesis by expanding our dataset to include more South Slavic languages, such as Serbian and Croatian. These languages are even closer to Slovene, because they belong to the same Western South Slavic subgroup and they use the Latin script. Serbian also has an official Cyrillic transliteration, which makes it perfect for further investigation of script significance.
- Since there is no official Slovene-to-Cyrillic transliteration standard, we followed Serbian transliteration rules—given its use of both scripts and linguistic proximity to Slovene—then adapted them to Bulgarian Cyrillic by replacing њ and љ with нь and љь. While our team includes native Cyrillic speakers, none are native in Slovene, so full accuracy cannot be guaranteed.

Appendix

A Model hyperparameters

Parameter	Value
Learning rate	3e-5
Batch size	16
Epochs	5
Weight decay	0.01
Warm-up ratio	0.1
Learning scheduler type	linear

Learning rate – standard for the type of our task

Epochs – only the final model with all four languages was trained on 3 epochs because of the restriction of our resources

Warm-up ratio – The use of warm-up steps is proven to lead to higher performance of Transformer models. This allows the learning rate to increase gradually and avoid poor convergence due to a rapid start.

B Contributions

We all equally contributed to all parts of the project.

C Use of chat-bots

- **Assistance purely with the language of the paper.** AI was used as a tool for writing corrections or for rephrasing some explanations.
- **Short-form input assistance.** There was no use of input assistance in our work.
- **Literature search.** We used AI to find information about previous works.
- **Low-novelty text.** We did not rely on AI for generating low-novelty text.
- **New ideas.** We did not rely on AI for getting new ideas.
- **New ideas + new text.** We used AI for finding and implementing lemmatization tools, but the idea originated from the Slav-NER papers. Every other idea was either implemented from scratch or adopted from previous works we found through manual search.

References

- Mikhail Arkhipov, Maria Trofimova, Yuri Kuratov, and Alexey Sorokin. 2019. [Tuning multilingual transformers for language-specific named entity recognition](#). In *Proceedings of the 7th Workshop on Balto-Slavic Natural Language Processing*, pages 89–93, Florence, Italy. Association for Computational Linguistics.
- Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. 2020. [Unsupervised cross-lingual representation learning at scale](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 8440–8451, Online. Association for Computational Linguistics.
- Pedro Ferreira, Ruben Cardoso, and Afonso Mendes. 2021. [Priberam labs at the 3rd shared task on SlavNER](#). In *Proceedings of the 8th Workshop on Balto-Slavic Natural Language Processing*, pages 86–92, Kiyv, Ukraine. Association for Computational Linguistics.
- Glottolog. 2024. [Family: Indo-european](#). Accessed: 2025-05-08.
- Chandler May, Ryan Cotterell, and Benjamin Van Durme. 2019. [An analysis of lemmatization on topic models of morphologically rich language](#). *Preprint*, arXiv:1608.03995.
- Rohan Paul. 2023. Zero shot learning multilingual-ner. https://github.com/rohan-paul/LLM-FineTuning-Large-Language-Models/blob/main/Other-Language_Models_BERT_related/Zero_Shot_Learning_multilingual-NER.ipynb. Accessed: 2025-05-15.
- Gabriela Pałka and Artur Nowakowski. 2023. [Exploring the use of foundation models for named entity recognition and lemmatization tasks in slavic languages](#). *ArXiv*, abs/2304.05336.
- Jakub Piskorski, Bogdan Babych, Zara Kancheva, Olga Kanishcheva, Maria Lebedeva, Michał Marcińczuk, Preslav Nakov, Petya Osenova, Lidia Pivovarova, Senja Pollak, Pavel Přibáň, Ivaylo Radev, Marko Robnik-Sikonja, Vasyl Starko, Josef Steinberger, and Roman Yangarber. 2021. [Slav-NER: the 3rd cross-lingual challenge on recognition, normalization, classification, and linking of named entities across Slavic languages](#). In *Proceedings of the 8th Workshop on Balto-Slavic Natural Language Processing*, pages 122–133, Kiyv, Ukraine. Association for Computational Linguistics.
- Jakub Piskorski, Michał Marcińczuk, and Roman Yangarber. 2024. [Cross-lingual named entity corpus for slavic languages](#). *Preprint*, arXiv:2404.00482.

- Matej Ulčar, Aleš Žagar, Carlos S. Armendariz, Andraž Repar, Senja Pollak, Matthew Purver, and Marko Robnik-Šikonja. 2021. [Evaluation of contextual embeddings on less-resourced languages](#). *Preprint*, arXiv:2107.10614.
- Rinalds Vīksna, Inguna Skadiņa, Daiga Deksne, and Roberts Rozis. 2023. [Large language models for multilingual Slavic named entity linking](#). In *Proceedings of the 9th Workshop on Slavic Natural Language Processing 2023 (SlavicNLP 2023)*, pages 172–178, Dubrovnik, Croatia. Association for Computational Linguistics.